Report: DNS Record Lookup (A, MX and TXT) for Three Real Domains

Tool: nslookup in Windows PowerShell | **Domains:** github.com, linkedin.com, google.com | **Resolver:** 1.1.1.1 (one.one.one.one)

1. Objective

To look up the A, MX and TXT records of three real domains with nslookup, record the output, and explain in plain language what each record type does.

2. Method

Each lookup used the record type switch, with the same resolver for all queries:

```
nslookup -type=A   <domain>
nslookup -type=MX  <domain>
nslookup -type=TXT <domain>
```

Every answer was marked **Non-authoritative**. That is normal: the answer came from the resolver 1.1.1.1 (its cache or its own lookup), not directly from the domain's own name server.

3. Results Summary

Domain	A record	MX records (preference, server)	TXT highlights
github.com	20.207.73.82	0, github-com.mail.protection.outlook.com	Verification tokens for Microsoft, Adobe, Apple, Atlassian, Calendly, Tailscale and others
linkedin.com	130.211.32.14	10, mail-a.linkedin.com; 10, mail-c.linkedin.com; 10, mail-d.linkedin.com; 20, mail.linkedin.com	Verification tokens for Apple, Atlassian, Bluebeam, DocuSign, plus AFDVALIDATION=LinkedIn
google.com	142.250.183.174	10, smtp.google.com	SPF policy (v=spf1 include:_spf.google.com ~all) plus many verification tokens

4. Detailed Results

4.1 github.com

A record. github.com resolves to the IPv4 address 20.207.73.82.

```
PS C:\Users\rishi> nslookup -type=A github.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:    github.com
Address:  20.207.73.82

PS C:\Users\rishi>
```

Figure 1: nslookup -type=A github.com.

MX record. One mail exchanger, with preference 0 (the highest possible priority): github-com.mail.protection.outlook.com. The name shows that GitHub's incoming email is handled by Microsoft's mail filtering service, not by servers GitHub runs itself.

```
PS C:\Users\rishi> nslookup -type=MX github.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
github.com      MX preference = 0, mail exchanger = github-com.mail.protection.outlook.com
PS C:\Users\rishi>
```

Figure 2: nslookup -type=MX github.com.

TXT records. The visible TXT records are mostly domain-verification strings, such as MS=..., adobe-idp-site-verification=..., apple-domain-verification=..., atlassian-domain-verification=..., calendly-site-verification=... and a TAILSCALE token. Each one proves to that company that GitHub controls the domain. The output continues beyond what the screenshot captured.

```
PS C:\Users\rishi> nslookup -type=TXT github.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
github.com      text =

        "00Dd0000000hHE0=1TBKg000000TN2r"
github.com      text =

        "MS=6BF03E6AF5CB689E315FB6199603BABF2C88D805"
github.com      text =

        "MS=ms44452932"
github.com      text =

        "MS=ms58704441"
github.com      text =

        "TAILSCALE-xOzoDvFUzZr5YYVCQFuD"
github.com      text =

        "adobe-idp-site-verification=b92c9e999aef825edc36e0a3d847d2dbad5b2fc0e05c79ddd7a16139b48ecf4b"
github.com      text =

        "anthropic-domain-verification-4az7qn=if8YWuRRqwLycGJDooumzHtxm"
github.com      text =

        "apple-domain-verification=RyQhdzTl6Z6x8ZP4"
github.com      text =

        "atlassian-domain-verification=jjgw98AKv2aeoYFxiL/VFaoyPkn3undEssTRuMg6C/3Fp/iqhkV4HVV7WjYlVeF8"
github.com      text =

        "calendly-site-verification=at0DQARi7IZvJtXQAWhMqpmIzpvoBNF7aam5VKKxP"
github.com      text =
```

Figure 3: nslookup -type=TXT github.com (first part of the output).

4.2 linkedin.com

A and MX records. linkedin.com resolves to 130.211.32.14. It has four MX records: three at preference 10 (mail-a, mail-c and mail-d) and one at preference 20 (mail.linkedin.com). The three equal-priority servers share the incoming mail load, and mail.linkedin.com is the backup used only if the others are unavailable. The line at the top of the figure appears to be leftover output from an earlier command and is not part of these results.

```
    timeout was 2 seconds.
*** Request to UnKnown timed-out
PS C:\Users\rishi> nslookup -type=MX  linkedin.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
linkedin.com    MX preference = 10, mail exchanger = mail-a.linkedin.com
linkedin.com    MX preference = 10, mail exchanger = mail-c.linkedin.com
linkedin.com    MX preference = 10, mail exchanger = mail-d.linkedin.com
linkedin.com    MX preference = 20, mail exchanger = mail.linkedin.com
PS C:\Users\rishi> nslookup -type=A  linkedin.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:    linkedin.com
Address:  130.211.32.14

PS C:\Users\rishi>
```

Figure 4: nslookup -type=MX and -type=A for linkedin.com.

TXT records. The visible records are verification strings: AFDVALIDATION=LinkedIn, apple-domain-verification, two atlassian-domain-verification entries, atlassian-sending-domain-verification, bluebeam-verification and docusign=, along with several plain token strings. The output continues beyond the screenshot.

```
Windows PowerShell
PS C:\Users\rishi> nslookup -type=TXT linkedin.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
linkedin.com    text =

        "448e0dc03e935ecf66d81f1ce3c26b2f2fea13756c031ffc4be91749107f3a79"
linkedin.com    text =

        "AFDVALIDATION=LinkedIn"
linkedin.com    text =

        "_cthqqp5zj8g86qf3h97heqitg2zc32b"
linkedin.com    text =

        "apple-domain-verification=Hp7LihDNsREwfHX9"
linkedin.com    text =

        "atlassian-domain-verification=dDed2VFvlDajBX8X22w52Jx/W/YLHR81SxUuraa9zNdz4aLjDS/RpfN11w2bxpRc"
linkedin.com    text =

        "atlassian-domain-verification=juKdSE4GGphSmzPkhmnRJUNIn0ALdb0vsP7VSPM8TmTP7WgbgUQPLFdNicP7bF58"
linkedin.com    text =

        "atlassian-sending-domain-verification=3bdb0597-814d-4e10-a552-5cf78f92ab3c"
linkedin.com    text =

        "bf5fl8sny79w4c70jxp6cp3crkqtr7qk"
linkedin.com    text =

        "bluebeam-verification=02px6k8snlrzd6gx3b3oqkeatbzawd"
linkedin.com    text =

        "docusign=11f01284-dffc-40f9-8d56-57e5261ede3f"
```

Figure 5: nslookup -type=TXT linkedin.com (first part of the output).

4.3 google.com

MX and TXT records. google.com has a single MX record, preference 10, pointing to smtp.google.com. The TXT output is long and contains many verification entries (onetrust, docusign, arcules, globalsign-smime-dv, google-site-verification, cisco-ci-domain-verification, facebook-domain-verification, apple-domain-verification and others).

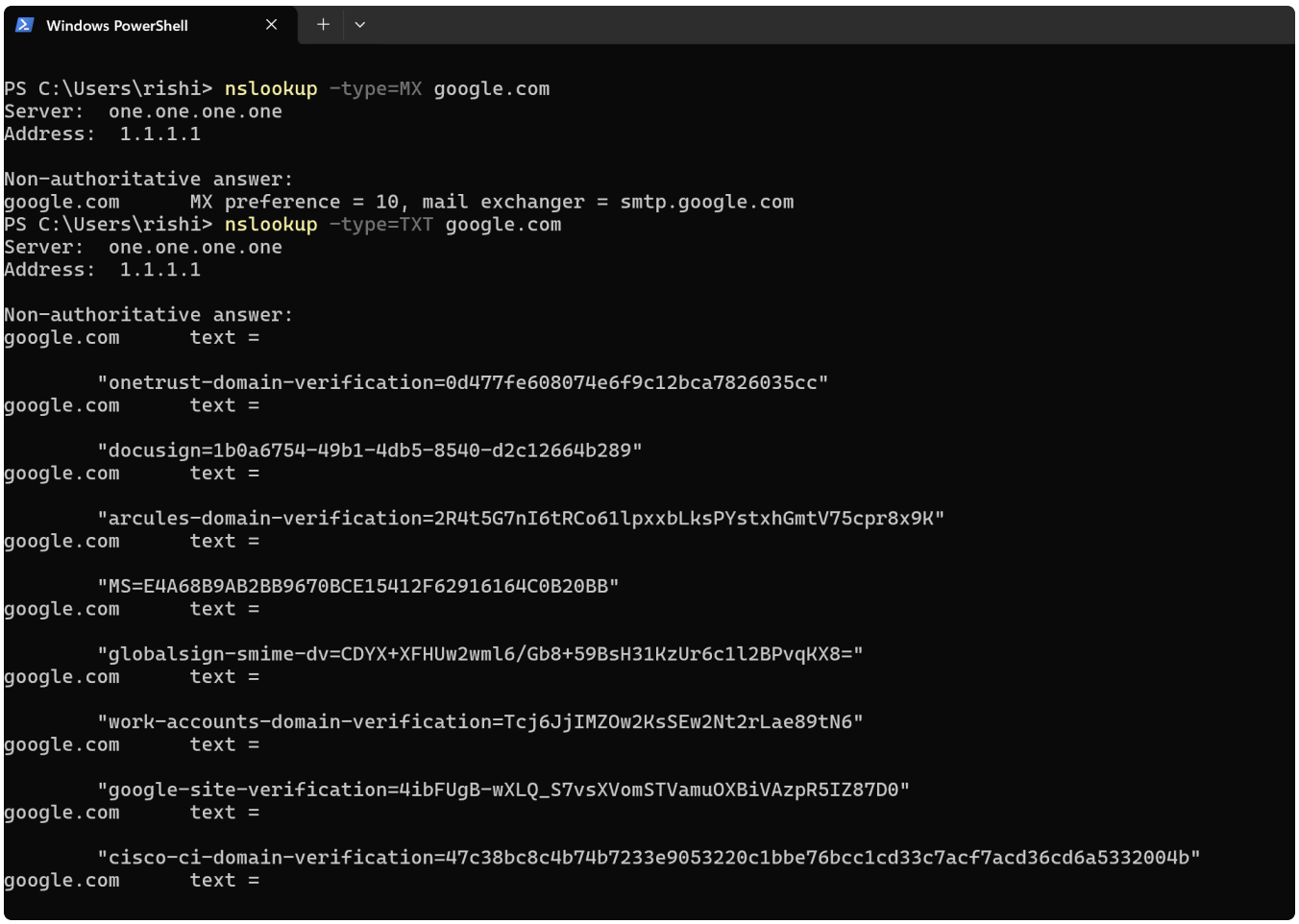


Figure 6: nslookup -type=MX and -type=TXT for google.com.

TXT (continued) and A record. The remaining TXT entries include the SPF policy `v=spf1 include:_spf.google.com ~all`. The A lookup shows google.com resolving to 142.250.183.174.



Figure 7: end of the google.com TXT output and nslookup -type=A google.com.

5. What Each Record Type Does

A record (Address). Translates a domain name into an IPv4 address, like looking up a name in a phone book to get a number. When you type github.com, your computer asks DNS for the A record and then connects to 20.207.73.82. (An AAAA record does the same job for IPv6.)

MX record (Mail Exchanger). Tells the world which servers accept email for a domain. When someone emails a person at linkedin.com, the sender's mail server looks up the MX records to find where to deliver it. Each record carries a **preference number**: the lowest number is tried first, and higher numbers are backups. Equal numbers, as with LinkedIn's three servers at 10, spread the load between them.

TXT record (Text). Stores free-form text in a domain's DNS. It does not direct web or mail traffic itself. It is used to prove or declare things about the domain:

- **Domain verification:** a service such as Atlassian, Apple or Google gives the owner a token to publish, which proves the owner controls the domain. Most of the TXT records in these screenshots are of this kind.

- **SPF (Sender Policy Framework):** lists which servers may send email for the domain, to help stop forged mail. Google's record `v=spf1 include:_spf.google.com ~all` means mail is legitimate if it comes from the servers listed in _spf.google.com, and anything else should be treated with suspicion (`~all` is a soft fail).

6. Observations

- **Different mail strategies.** GitHub uses an outsourced mail service (Microsoft), LinkedIn runs its own multiple servers with a backup, and Google uses a single hostname.
- **Priority numbers differ.** GitHub's 0, Google's 10 and LinkedIn's 10 and 20 all work the same way; only the order matters, not the absolute value.
- **TXT records pile up.** Large organizations accumulate many verification tokens over the years as they adopt different services.
- **Results can change.** Large sites often return different IP addresses by location or over time, so another run may show different A records.

7. Conclusion

nslookup showed how one domain name serves several purposes in DNS: the A record points visitors to a web server, the MX records direct incoming email to the right mail servers in priority order, and the TXT records carry ownership proofs and email security policy. Together they show how DNS underpins both web access and email delivery.